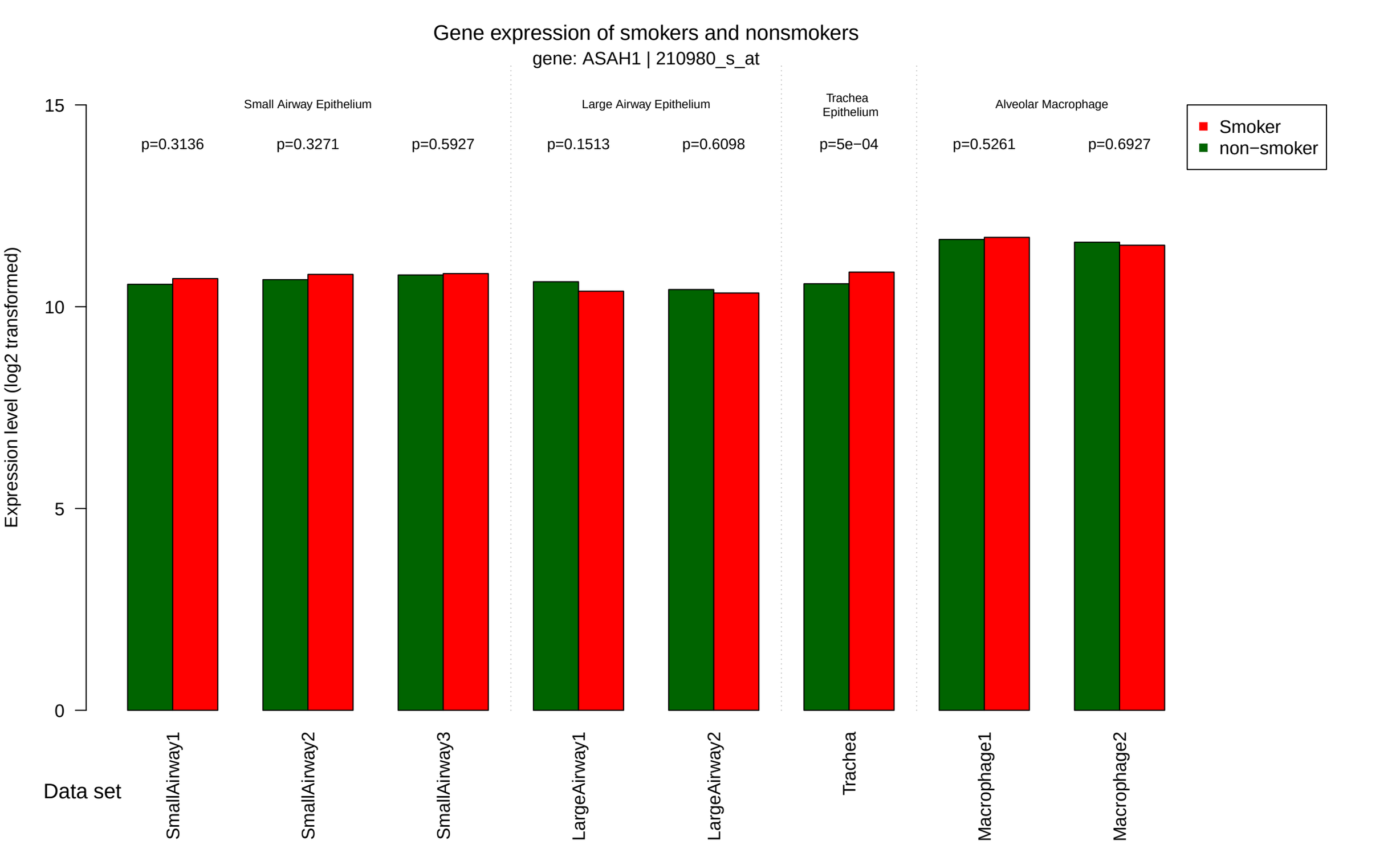
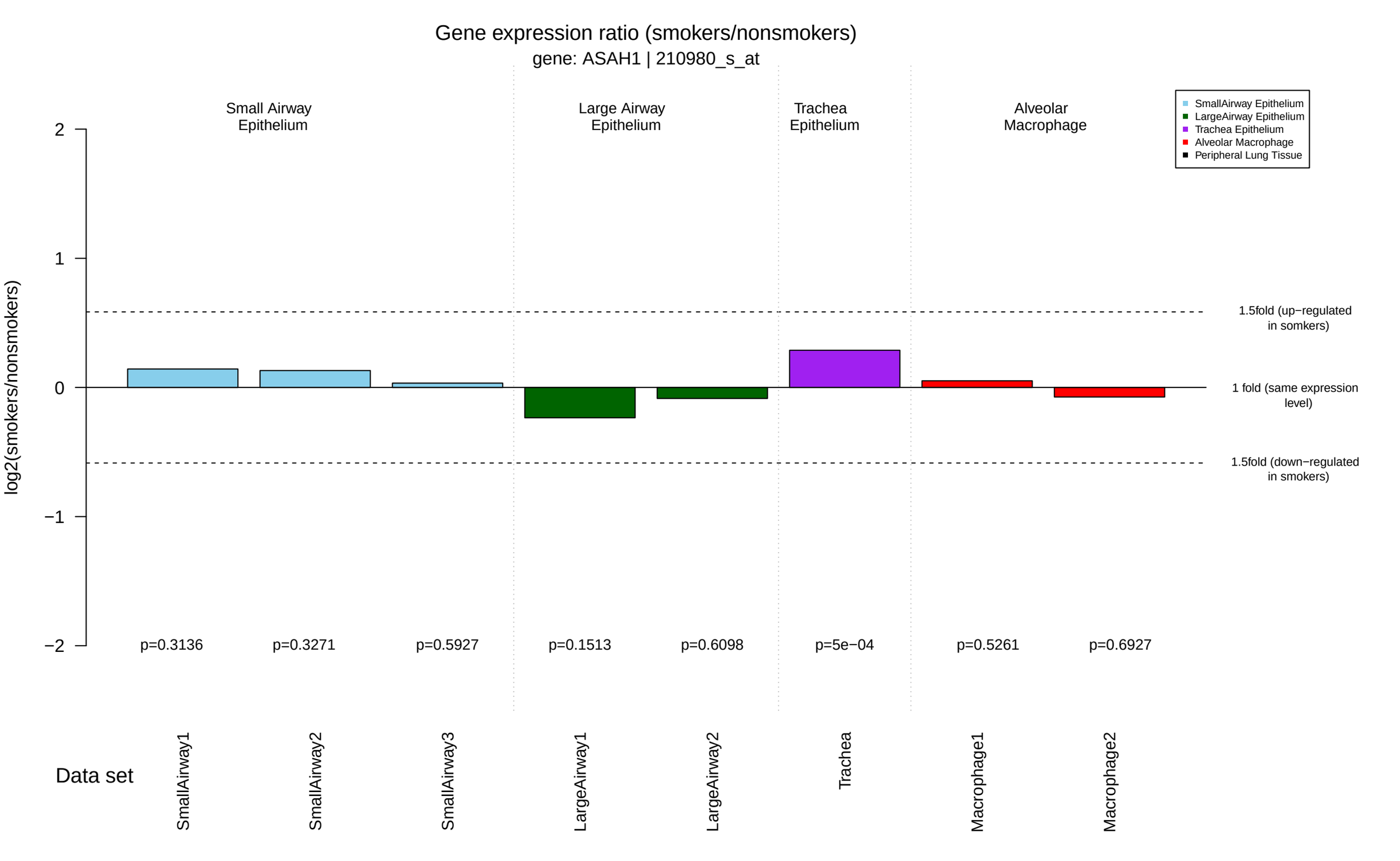
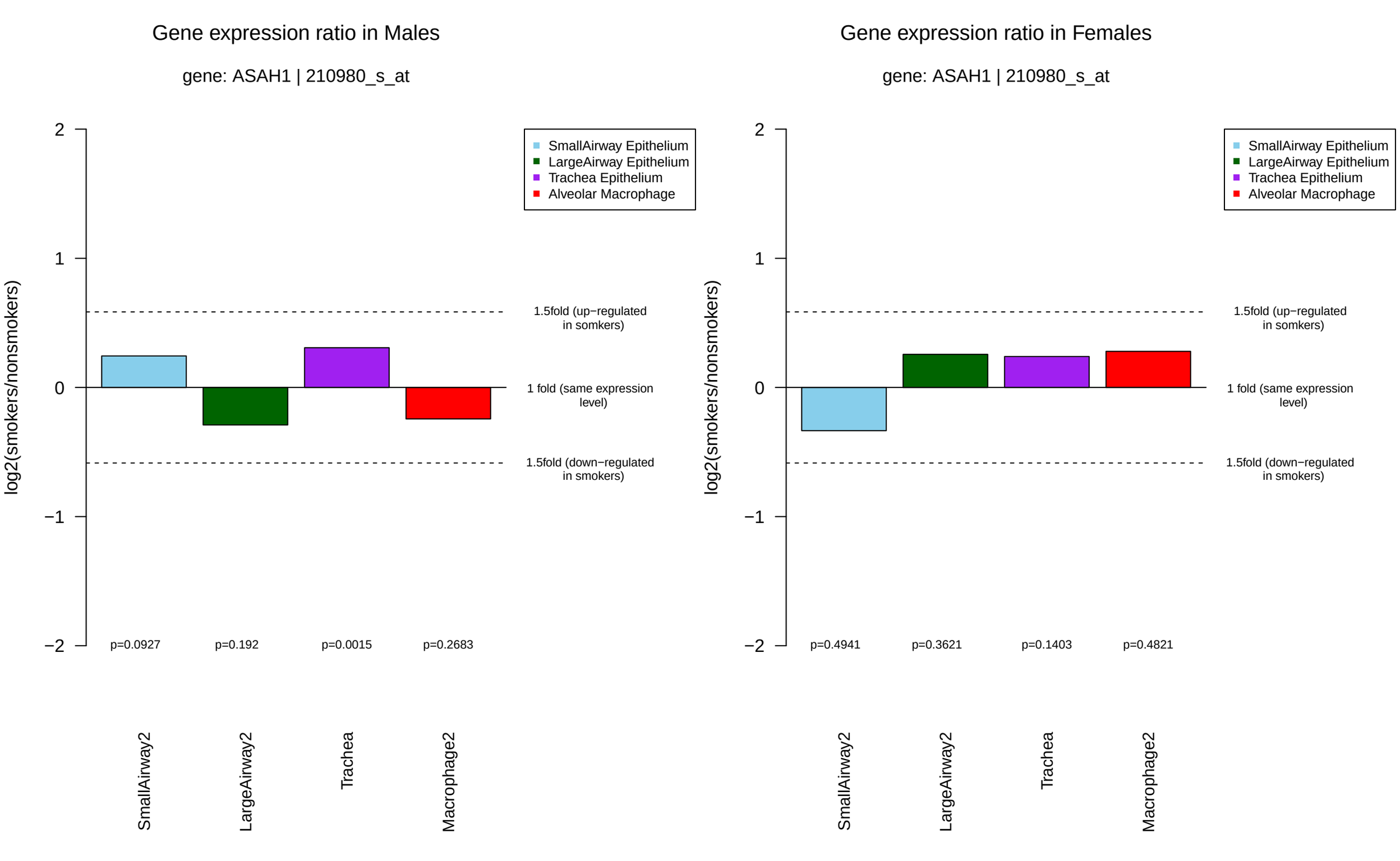
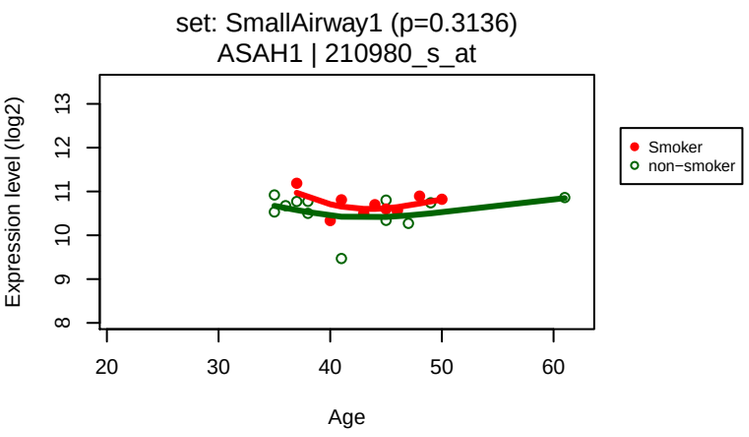
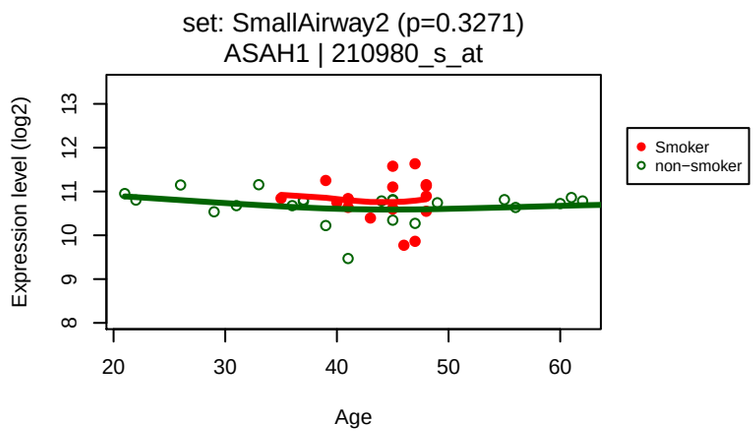
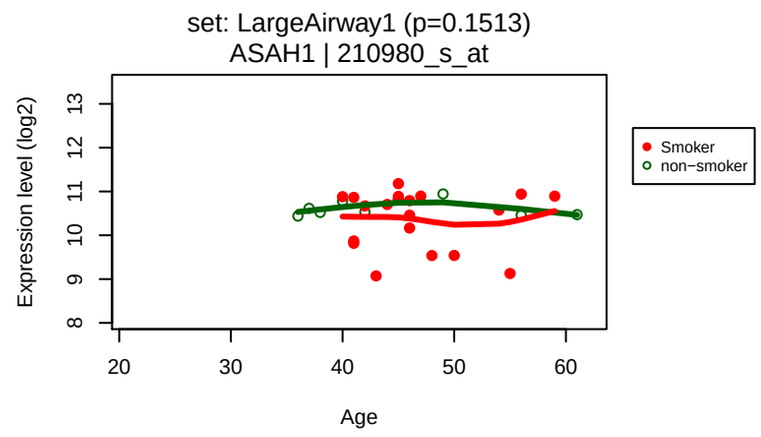

Gene expression of smokers and nonsmokers
gene: ASAH1 | 210980_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3136
p=0.3271
p=0.5927
p=0.1513
p=0.6098
p=5e−04
p=0.5261
p=0.6927
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

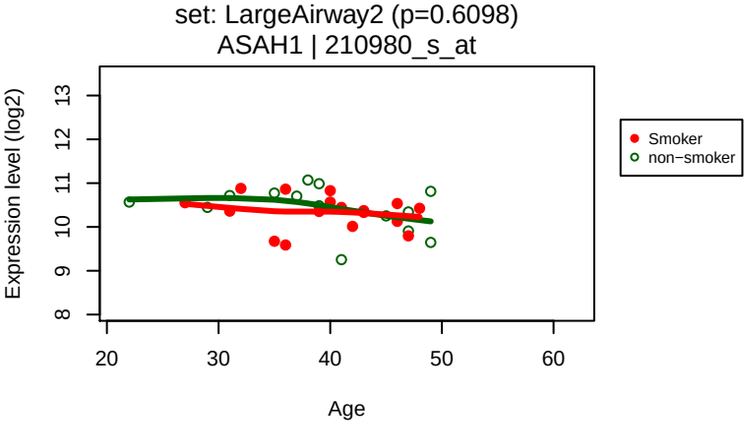

Gene expression ratio (smokers/nonsmokers)
gene: ASAH1 | 210980_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3136
p=0.3271
p=0.5927
p=0.1513
p=0.6098
p=5e−04
p=0.5261
p=0.6927
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

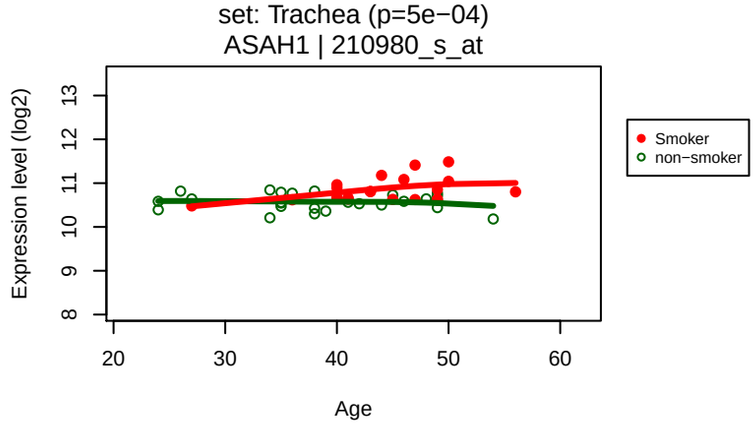

Gene expression ratio in Males
gene: ASAH1 | 210980_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0927
p=0.192
p=0.0015
p=0.2683
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ASAH1 | 210980_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4941
p=0.3621
p=0.1403
p=0.4821
SmallAirway2
LargeAirway2
Trachea
Macrophage2

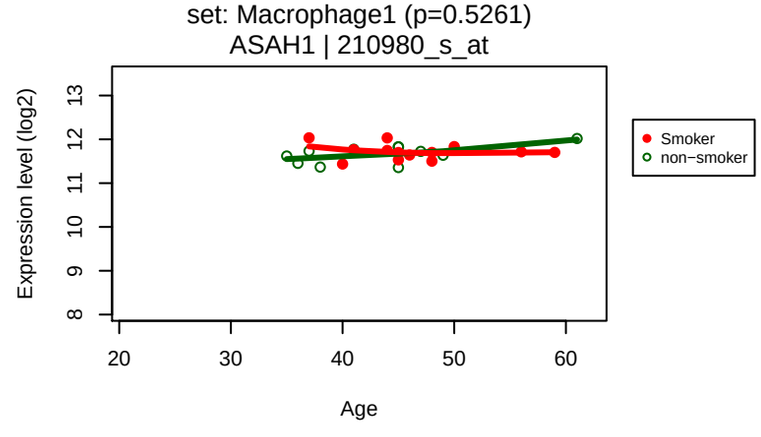

set: SmallAirway1 (p=0.3136)
ASAH1 | 210980_s_at
Expression level (log2)
Age
Smoker
non−smoker

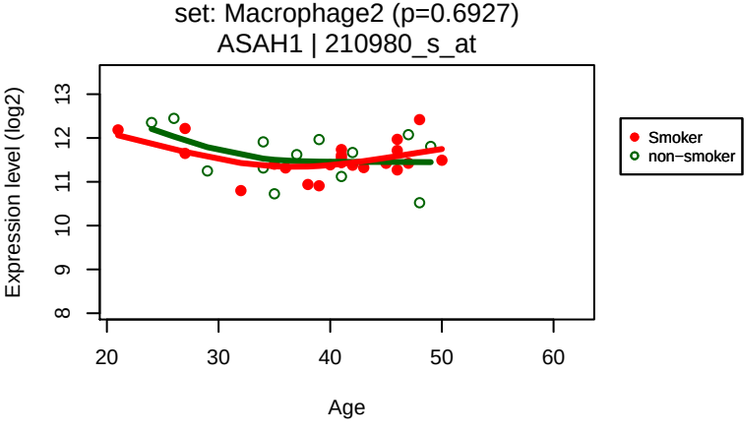

set: SmallAirway2 (p=0.3271)
ASAH1 | 210980_s_at
Expression level (log2)
Age
Smoker
non−smoker

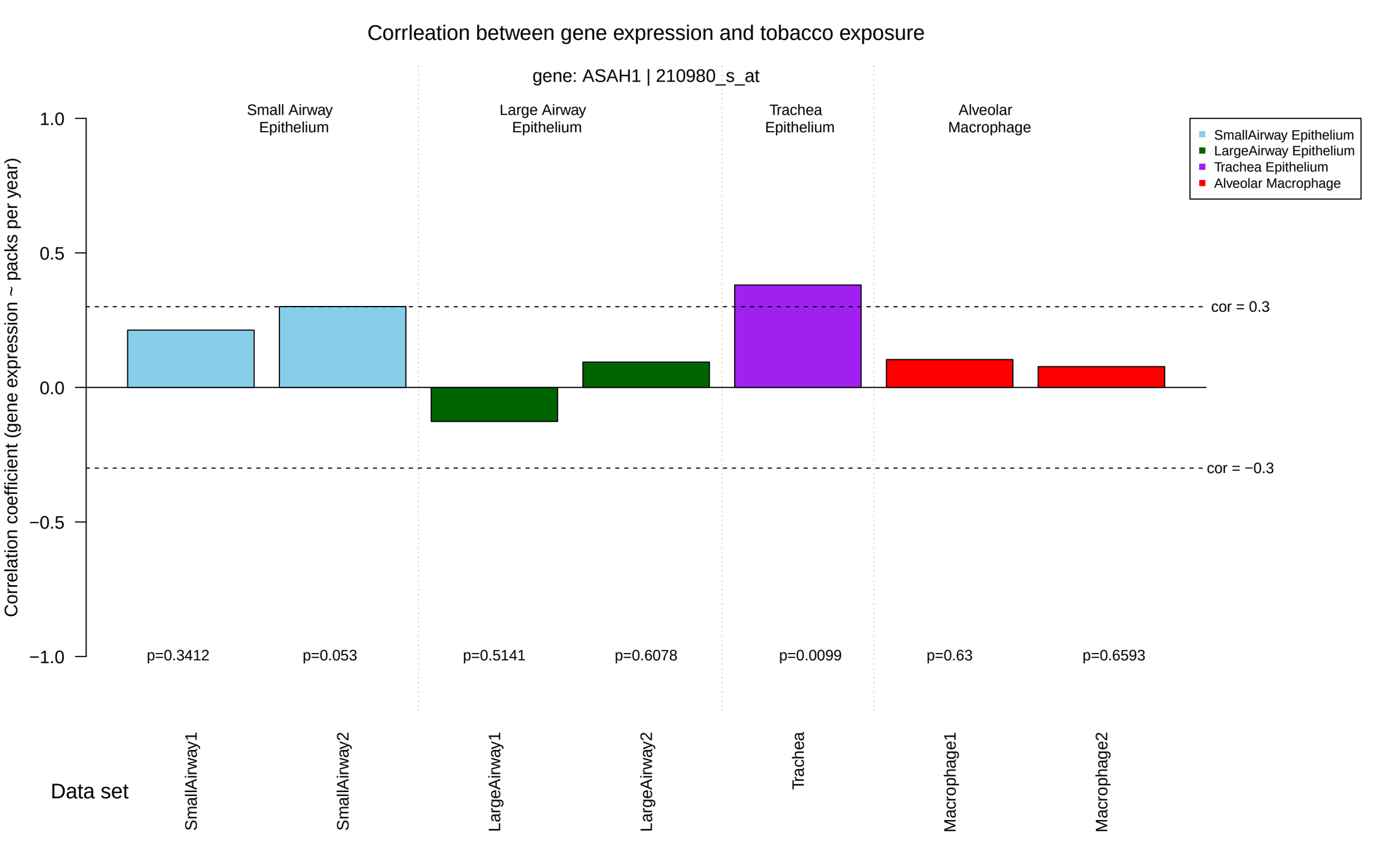

set: LargeAirway1 (p=0.1513)
ASAH1 | 210980_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6098)
ASAH1 | 210980_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=5e−04)
ASAH1 | 210980_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5261)
ASAH1 | 210980_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6927)
ASAH1 | 210980_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ASAH1 | 210980_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3412
p=0.053
p=0.5141
p=0.6078
p=0.0099
p=0.63
p=0.6593
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2